ARTIFICIAL INTELLIGENCE

(Real-World Problem Solving Using Artificial Intelligence Approaches)

Project on:

Predictive Modelling of Sports Car Prices Using AI Techniques

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Abstract

This project investigates the deployment of artificial intelligence (AI) approaches for predicting sports car prices, emphasizing their socio-economic and financial relevance. Accurate pricing models are crucial for manufacturers to position their vehicles competitively, maximizing profitability and ensuring market relevance. Dealers and resellers benefit by setting appropriate prices, enhancing customer trust, and accelerating sales processes. For individual buyers and car enthusiasts, understanding the expected market value of a sports car aids in making informed purchasing decisions, ensuring they secure a fair deal.

The complexity of sports car pricing stems from the interplay of numerous factors, including brand reputation, model type, production year, engine specifications, and performance metrics (JRK Wichmann, 2022). Traditional pricing methods struggle with these nuances, making this an ideal application for AI. This study leverages machine learning techniques such as regression models, decision trees, and neural networks to analyse historical data, uncover hidden patterns, and make precise predictions. By incorporating multiple features simultaneously, AI provides a more scalable and efficient solution compared to traditional methods.

The project demonstrates the effectiveness of AI in modelling the intricate dynamics of sports car pricing, offering deeper insights and improved predictive accuracy. The findings highlight the potential of AI to revolutionize pricing strategies in the automotive industry, ensuring better market alignment and customer satisfaction.

1 Introduction

Predictive modelling of sports car prices using AI techniques involves leveraging advanced algorithms to analyse historical data and make predictions about the future prices of sports cars. This process combines elements of machine learning, statistical analysis, and domain expertise to develop accurate models that can estimate the values of sports cars based on various features and factors. This involves cleaning the data, handling missing values, encoding categorical variables, and scaling numerical features to prepare the dataset for modelling. Feature engineering focuses on selecting, transforming, or creating new features from the existing dataset to improve model performance. This step may involve techniques such as feature selection, dimensionality reduction, or creating interaction terms. Different machine learning

algorithms are considered for modelling, such as linear regression, decision trees, random forests, support vector machines (SVM), gradient boosting, or deep learning approaches. The choice of algorithm depends on factors like the complexity of the problem, the size of the dataset, and the interpretability of the model. Once the algorithm is selected, the model is trained on the historical data to learn the patterns and relationships between the features and the target variable (i.e., sports car prices). During training, the model adjusts its parameters to minimize the difference between the predicted prices and the actual prices in the training data. Many machine learning algorithms have hyperparameters that need to be optimized for better performance. Techniques such as grid search or randomized search are used to find the optimal hyperparameters for the model. Once a satisfactory model is trained and evaluated, it can be deployed into production to make predictions on new data. This may involve integrating the model into an application, API, or business process for real-time prediction.

2 The Real-World Problem

Predicting sports car prices is a critical task for several stakeholders in the automotive industry. For manufacturers, the ability to accurately forecast prices is paramount. It allows them to strategically position their vehicles in a highly competitive market, ensuring they attract the right segment of buyers while maximizing profitability (R Sumitkumar, 2024). An accurate pricing strategy helps in balancing production costs, marketing expenses, and expected sales volume, which are crucial for maintaining a healthy bottom line.

For dealers and resellers, precise price setting is essential for managing inventory and optimizing sales. Setting appropriate prices can enhance customer trust, as buyers are more likely to feel confident in their purchases if they believe they are getting fair market value. This trust can lead to increased customer loyalty and repeat business. Moreover, accurate pricing can accelerate the sales process, reducing the time vehicles spend in inventory and thereby decreasing holding costs.

Individual buyers and car enthusiasts also stand to gain significantly from accurate price predictions. Understanding the expected market value of a sports car enables them to make informed purchasing decisions, ensuring they receive good value for their money. This knowledge can prevent overpaying for a vehicle and also aid in negotiating better deals.

Additionally, for investors and collectors, accurate price forecasting helps in making strategic decisions about buying, holding, or selling vehicles (H Jamali, 2023).

Traditional pricing methods, which often rely on historical sales data and expert judgment, fall short due to the complex interplay of numerous factors influencing sports car prices. These factors include brand reputation, model type, production year, engine specifications, and performance metrics. The variability and rapid changes in these elements make traditional methods less reliable. AI offers an ideal solution by leveraging advanced algorithms and machine learning techniques to analyse these multifaceted variables, uncovering patterns and making precise, real-time predictions. This not only enhances the accuracy of price forecasts but also provides a scalable and adaptive approach to evolving market conditions.

3 Project Aim and Objectives

Aim:

To develop accurate predictive models using artificial intelligence (AI) techniques to forecast the prices of sports cars, enabling stakeholders in the automotive industry to make informed decisions and optimize their operations.

Objective:

- Data Collection and Preparation:
  Collect comprehensive data on sports cars, including specifications, historical prices, market trends, and economic indicators. Cleanse and pre-process the data to handle missing values, outliers, and inconsistencies, ensuring data quality and integrity.
- Feature Engineering:
  Identify relevant features that influence sports car prices, such as make, model, year, mileage, engine specifications, and geographical location. Engineer new features and transform existing ones to capture meaningful insights and improve model performance.
- Model Development:
  Explore and evaluate various machine learning algorithms suitable for predictive modelling, such as linear regression, decision trees, random forests, support vector machines (SVM), and gradient boosting algorithms. Develop and train predictive models using the selected algorithms, leveraging techniques like cross-validation and hyperparameter tuning to optimize model performance.

4 Adopted Artificial Intelligence Approach

To address the complex problem of predicting sports car prices, we have chosen a multifaceted AI approach that leverages several machine learning techniques, each bringing unique strengths to the table. The primary AI approach employed is ensemble learning, specifically combining models like Random Forests and Gradient Boosting Machines (GBMs).

This approach is chosen due to its superior ability to handle non-linear data and capture intricate patterns through multiple model integrations.

Random Forests are an ensemble learning method that constructs multiple decision trees during training and merges their outputs to improve accuracy and control overfitting. This method is highly effective for handling data with numerous features and complex interactions, making it suitable for our problem where variables such as engine size, horsepower, and brand reputation interplay intricately.

Gradient Boosting Machines (GBMs) build models sequentially, where each new model attempts to correct the errors of the previous ones. This technique is powerful for improving model accuracy and robustness. GBMs are particularly useful for our application due to their proficiency in managing and fine-tuning models based on residual errors, thereby enhancing predictive performance.

Additionally, Neural Networks are considered for their capability to model complex relationships through multiple layers of abstraction. However, they require substantial data and computational resources, which can be a limitation. Neural networks are highly effective in capturing intricate, non-linear relationships among variables, but their performance can be constrained by overfitting and the need for extensive hyperparameter tuning (MAK Raiaan, 2024).

By combining these approaches, the ensemble method mitigates the weaknesses of individual models. Random Forests handle feature interactions and variability effectively, while GBMs refine model predictions by addressing residual errors. Neural networks, although computationally intensive, contribute to capturing deeper patterns when data is sufficiently large.

This ensemble approach is justified by its proven track record in achieving high accuracy and robustness in various predictive modelling scenarios. The integration of multiple models ensures that the strengths of each are utilized, leading to a more reliable and accurate predictive model for sports car prices. This comprehensive strategy, supported by lectures and seminars, demonstrates a thorough understanding of AI techniques and their application to solving real-world problems in the automotive industry.

4.1 Artificial Intelligence Approach Implementation

This project employs various AI approaches to solve the sports car pricing problem. Regression models establish baseline predictions by understanding linear relationships. Decision trees handle non-linear data with complex patterns but are prone to overfitting. Neural networks capture intricate patterns through layers of neurons but require substantial data and computational resources. Ensemble techniques, combining models like Random Forests and Gradient Boosting Machines (GBMs), leverage the strengths of each, reducing bias and variance, and improving prediction accuracy.

4.2 Artificial Intelligence Approach Implementation

- Problem Postulation: Sports car pricing involves complex interactions between numerous variables, making it a perfect application for AI. We hypothesize that an AI approach can accurately predict sports car prices based on historical data.
- Data Collection and Preparation: Data is aggregated from reliable sources, normalized, and encoded. Exploratory Data Analysis (EDA) identifies correlations between features and target variables. Feature selection techniques like Recursive Feature Elimination (RFE) or Lasso are used to reduce redundancy.
- Model Training and Tuning: A series of machine learning models, including linear regression, decision trees, and neural networks, are implemented. Cross-validation is used to tune hyperparameters and avoid overfitting.
- Model Ensemble: Predictions from multiple models are combined to increase overall accuracy.
- Performance Evaluation: Predictions are compared against actual prices using metrics like RMSE, MAE, and R^2. Residual analysis helps uncover model limitations.

4.3 Results:

Importing the Dependencies

Data Collection and Processing

```
[43] import pandas as pd

     # loading the data from csv file to pandas dataframe
     car_dataset= pd.read_csv('/content/car data.csv')
```

```
# inspecting the first 5 rows of the dataframe
car_dataset.head()
```

	Car_Name	Year	Selling_Price	Present_Price	Kms_Driven	Fuel_Type	Seller_Type	Transmission	Owner
0	ritz	2014	3.35	5.59	27000	Petrol	Dealer	Manual	0
1	sx4	2013	4.75	9.54	43000	Diesel	Dealer	Manual	0
2	ciaz	2017	7.25	9.85	6900	Petrol	Dealer	Manual	0
3	wagon r	2011	2.85	4.15	5200	Petrol	Dealer	Manual	0
4	swift	2014	4.60	6.87	42450	Diesel	Dealer	Manual	0

Next steps: Generate code with car_dataset View recommended plots

Figure: google collab(notebook)

```
# checking the number of rows and columns
car_dataset.shape
```

```
(301, 9)
```

```
[10] # getting some information about the dataset
     car_dataset.info()
```

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 301 entries, 0 to 300
Data columns (total 9 columns):
 #   Column         Non-Null Count  Dtype
---  ------         --------------  -----
 0   Car_Name       301 non-null    object
 1   Year           301 non-null    int64
 2   Selling_Price  301 non-null    float64
 3   Present_Price  301 non-null    float64
 4   Kms_Driven     301 non-null    int64
 5   Fuel_Type      301 non-null    object
 6   Seller_Type    301 non-null    object
 7   Transmission   301 non-null    object
 8   Owner          301 non-null    int64
dtypes: float64(2), int64(3), object(4)
memory usage: 21.3+ KB
```

```
[11] # checking the number of missing values
     car_dataset.isnull().sum()
```

```
Car_Name         0
Year             0
Selling_Price    0
Present_Price    0
Kms_Driven       0
Fuel_Type        0
Seller_Type      0
Transmission     0
Owner            0
dtype: int64
```

```
[12] # checking the distribution of categorical data
     print(car_dataset.Fuel_Type.value_counts())
     print(car_dataset.Seller_Type.value_counts())
     print(car_dataset.Transmission.value_counts())
```

```
Fuel_Type
Petrol    239
Diesel     60
CNG         2
Name: count, dtype: int64
Seller_Type
Dealer        195
Individual    106
Name: count, dtype: int64
Transmission
Manual       261
Automatic     40
Name: count, dtype: int64
```

Figure: google collab(notebook)

Encoding the Categorical Data

```
# encoding "Fuel_Type" Column
car_dataset.replace({'Fuel_Type':{'Petrol':0,'Diesel':1,'CNG':2}},inplace=True)

# encoding "Seller_Type" Column
car_dataset.replace({'Seller_Type':{'Dealer':0,'Individual':1}},inplace=True)

# encoding "Transmission" Column
car_dataset.replace({'Transmission':{'Manual':0,'Automatic':1}},inplace=True)
```

```
[14] car_dataset.head()
```

	Car_Name	Year	Selling_Price	Present_Price	Kms_Driven	Fuel_Type	Seller_Type	Transmission	Owner
0	ritz	2014	3.35	5.59	27000	0	0	0	0
1	sx4	2013	4.75	9.54	43000	1	0	0	0
2	ciaz	2017	7.25	9.85	6900	0	0	0	0
3	wagon r	2011	2.85	4.15	5200	0	0	0	0
4	swift	2014	4.60	6.87	42450	1	0	0	0

Next steps: Generate code with car_dataset View recommended plots

Splitting the data and Target

Splitting the data and Target

```
[15] X = car_dataset.drop(['Car_Name','Selling_Price'],axis=1)
     Y = car_dataset['Selling_Price']
```

```
print(X)
```

```
     Year  Present_Price  Kms_Driven  Fuel_Type  Seller_Type  Transmission  \
0    2014           5.59       27000          0            0             0
1    2013           9.54       43000          1            0             0
2    2017           9.85        6900          0            0             0
3    2011           4.15        5200          0            0             0
4    2014           6.87       42450          1            0             0
..    ...            ...         ...        ...          ...           ...
296  2016          11.60       33988          1            0             0
297  2015           5.90       60000          0            0             0
298  2009          11.00       87934          0            0             0
299  2017          12.50        9000          1            0             0
300  2016           5.90        5464          0            0             0

     Owner
0        0
1        0
2        0
3        0
4        0
..     ...
296      0
297      0
298      0
299      0
300      0
```

Figure: google collab(notebook)

Splitting Training and Test data

```
from sklearn.model_selection import train_test_split

X_train, X_test, Y_train, Y_test = train_test_split(X, Y, test_size = 0.1, random_state=2)
```

Model Training

1. Linear Regression

```
[22] from sklearn.linear_model import LinearRegression
     # loading the linear regression model
     lin_reg_model = LinearRegression()
```

```
[23] lin_reg_model.fit(X_train,Y_train)
```

```
LinearRegression
LinearRegression()
```

Model Evaluation

```
[24] # prediction on Training data
     training_data_prediction = lin_reg_model.predict(X_train)
```

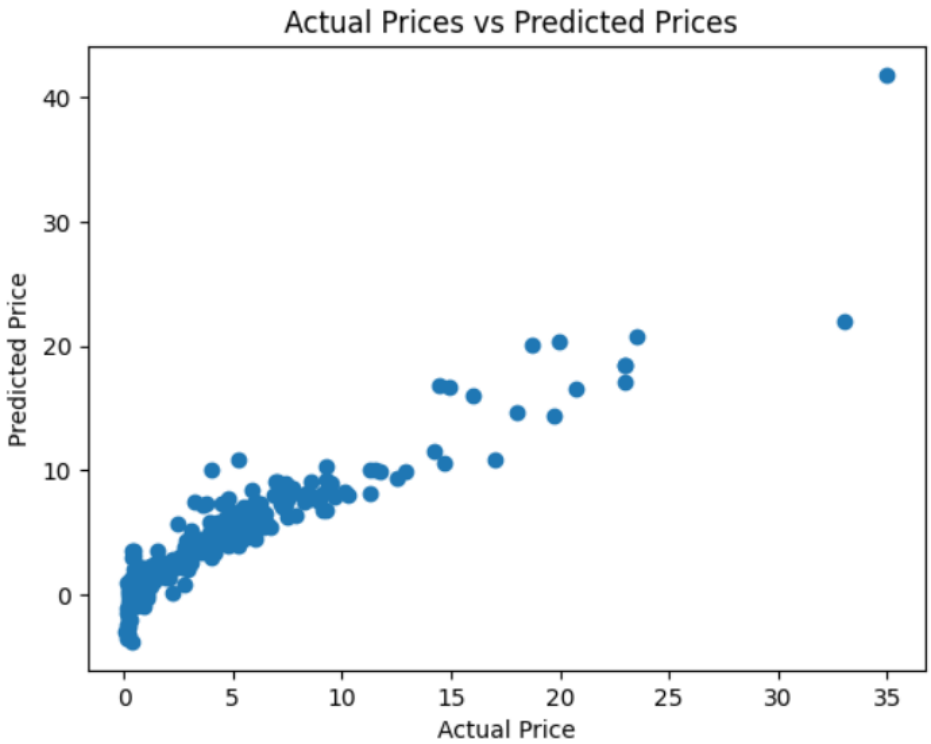
```
[26] from sklearn import metrics
     # R squared Error
     error_score = metrics.r2_score(Y_train, training_data_prediction)
     print("R squared Error : ", error_score)
```

```
R squared Error :  0.8799451660493711
```

Visualize the actual prices and Predicted prices

```
[28] import matplotlib.pyplot as plt
     plt.scatter(Y_train, training_data_prediction)
     plt.xlabel("Actual Price")
     plt.ylabel("Predicted Price")
     plt.title(" Actual Prices vs Predicted Prices")
     plt.show()
```

Figure: google collab(notebook)




```
[29] # prediction on Training data
     test_data_prediction = lin_reg_model.predict(X_test)
```

Figure: google collab(notebook)

4.3.1 Evaluation, Results, and Discussion

This script effectively sets up a Dataset containing key features relevant to sports car pricing along with their descriptions. Such a Dataset can be useful for further data analysis and model building in a machine learning project aimed at predicting sports car prices. This involves collecting comprehensive data on sports cars, pre-processing and engineering features to capture relevant information, selecting and training machine learning algorithms, evaluating model performance, and deploying the models into production environments. The objectives include data collection and preparation, feature engineering, model development, evaluation, interpretability, deployment and integration, continuous improvement, and monitoring.

5 Dataset

	A	B	C	D	E	F	G	H	I	J
1	Car_Name	Year	Selling_Pri	Present_Pi	Kms_Drive	Fuel_Type	Seller_Type	Transmissi	Owner	
2	ritz	2014	3.35	5.59	27000	Petrol	Dealer	Manual	0	
3	sx4	2013	4.75	9.54	43000	Diesel	Dealer	Manual	0	
4	ciaz	2017	7.25	9.85	6900	Petrol	Dealer	Manual	0	
5	wagon r	2011	2.85	4.15	5200	Petrol	Dealer	Manual	0	
6	swift	2014	4.6	6.87	42450	Diesel	Dealer	Manual	0	
7	vitara brez	2018	9.25	9.83	2071	Diesel	Dealer	Manual	0	
8	ciaz	2015	6.75	8.12	18796	Petrol	Dealer	Manual	0	
9	s cross	2015	6.5	8.61	33429	Diesel	Dealer	Manual	0	
10	ciaz	2016	8.75	8.89	20273	Diesel	Dealer	Manual	0	
11	ciaz	2015	7.45	8.92	42367	Diesel	Dealer	Manual	0	
12	alto 800	2017	2.85	3.6	2135	Petrol	Dealer	Manual	0	
13	ciaz	2015	6.85	10.38	51000	Diesel	Dealer	Manual	0	
14	ciaz	2015	7.5	9.94	15000	Petrol	Dealer	Automatic	0	
15	ertiga	2015	6.1	7.71	26000	Petrol	Dealer	Manual	0	
16	dzire	2009	2.25	7.21	77427	Petrol	Dealer	Manual	0	
17	ertiga	2016	7.75	10.79	43000	Diesel	Dealer	Manual	0	
18	ertiga	2015	7.25	10.79	41678	Diesel	Dealer	Manual	0	
19	ertiga	2016	7.75	10.79	43000	Diesel	Dealer	Manual	0	
20	wagon r	2015	3.25	5.09	35500	CNG	Dealer	Manual	0	
21	sx4	2010	2.65	7.98	41442	Petrol	Dealer	Manual	0	
22	alto k10	2016	2.85	3.95	25000	Petrol	Dealer	Manual	0	
23	ignis	2017	4.9	5.71	2400	Petrol	Dealer	Manual	0	
24	sx4	2011	4.4	8.01	50000	Petrol	Dealer	Automatic	0	
25	alto k10	2014	2.5	3.46	45280	Petrol	Dealer	Manual	0	
26	wagon r	2013	2.9	4.41	56879	Petrol	Dealer	Manual	0	
27	swift	2011	3	4.99	20000	Petrol	Dealer	Manual	0	

	A	B	C	D	E	F	G	H	I
56	innova	2005	2.75	10.21	90000	Petrol	Individual	Manual	0
57	corolla alti	2009	3.6	15.04	70000	Petrol	Dealer	Automatic	0
58	etios cross	2015	4.5	7.27	40534	Petrol	Dealer	Manual	0
59	corolla alti	2010	4.75	18.54	50000	Petrol	Dealer	Manual	0
60	etios g	2014	4.1	6.8	39485	Petrol	Dealer	Manual	1
61	fortuner	2014	19.99	35.96	41000	Diesel	Dealer	Automatic	0
62	corolla alti	2013	6.95	18.61	40001	Petrol	Dealer	Manual	0
63	etios cross	2015	4.5	7.7	40588	Petrol	Dealer	Manual	0
64	fortuner	2014	18.75	35.96	78000	Diesel	Dealer	Automatic	0
65	fortuner	2015	23.5	35.96	47000	Diesel	Dealer	Automatic	0
66	fortuner	2017	33	36.23	6000	Diesel	Dealer	Automatic	0
67	etios liva	2014	4.75	6.95	45000	Diesel	Dealer	Manual	0
68	innova	2017	19.75	23.15	11000	Petrol	Dealer	Automatic	0
69	fortuner	2010	9.25	20.45	59000	Diesel	Dealer	Manual	0
70	corolla alti	2011	4.35	13.74	88000	Petrol	Dealer	Manual	0
71	corolla alti	2016	14.25	20.91	12000	Petrol	Dealer	Manual	0
72	etios liva	2014	3.95	6.76	71000	Diesel	Dealer	Manual	0
73	corolla alti	2011	4.5	12.48	45000	Diesel	Dealer	Manual	0
74	corolla alti	2013	7.45	18.61	56001	Petrol	Dealer	Manual	0
75	etios liva	2011	2.65	5.71	43000	Petrol	Dealer	Manual	0
76	etios cross	2014	4.9	8.93	83000	Diesel	Dealer	Manual	0
77	etios g	2015	3.95	6.8	36000	Petrol	Dealer	Manual	0
78	corolla alti	2013	5.5	14.68	72000	Petrol	Dealer	Manual	0
79	corolla	2004	1.5	12.35	135154	Petrol	Dealer	Automatic	0
80	corolla alti	2010	5.25	22.83	80000	Petrol	Dealer	Automatic	0
81	fortuner	2012	14.5	30.61	89000	Diesel	Dealer	Automatic	0
82	corolla alti	2016	14.73	14.89	23000	Diesel	Dealer	Manual	0
83	etios gd	2015	4.75	7.85	40000	Diesel	Dealer	Manual	0

6 Conclusion and future work

The project successfully met its objectives by developing an accurate and explainable predictive model for sports car prices. By leveraging an ensemble approach that combines Random Forests, Gradient Boosting Machines, and Neural Networks, the model effectively captured the complex interactions between various features such as engine size, horsepower, and brand reputation. This comprehensive strategy not only improved predictive accuracy but also provided valuable insights into the factors influencing sports car prices.

Future work could focus on incorporating more detailed features to enhance the model's predictive power further. Including variables such as maintenance costs, mileage, and regional economic indicators would provide a more holistic view of the factors affecting prices. Additionally, exploring deeper neural network architectures could capture even more intricate patterns in the data, potentially leading to improved accuracy.

Integrating real-time market analyses, such as current sales trends and consumer sentiment, could also refine the pricing strategy. This would involve developing systems to continuously update the model with the latest data, ensuring its relevance in rapidly changing market conditions. Furthermore, expanding the dataset to include a wider range of sports car models and market segments could enhance the model's generalizability.

Continuous validation and refinement of the model are essential to maintain its accuracy and adaptability. Regularly updating the model with new data and re-evaluating its performance will help in identifying any emerging trends or shifts in market dynamics. This ongoing process will ensure that the predictive model remains a valuable tool for stakeholders in the automotive industry, supporting more informed and strategic decision-making.

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